DATA SCIENCE
EXPERT ROADMAP

End-to-End Learning Path from Beginner to Industry Expert

2025 Edition • Original Curriculum Design

Covers: Mathematics · Programming · ML · Deep Learning · MLOps · Leadership

Designed for learners at every stage — from zero experience to senior-level expertise.

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1. INTRODUCTION & WHAT IS DATA SCIENCE?

What Is Data Science?

Data Science is an interdisciplinary practice that extracts actionable knowledge and insight from structured and unstructured data by combining domain expertise, programming skills, statistical methods, and machine learning. A Data Scientist turns raw data into decisions, products, and measurable business impact.

Why Become a Data Science Expert?

- Among the highest-paid technology roles globally, with median salaries exceeding $120,000 USD.
- Consistently ranked one of the most in-demand professions across industries.
- Applies across healthcare, finance, e-commerce, climate science, government, and every other field.
- Empowers you to solve complex real-world problems with a blend of creativity and rigour.
- Offers diverse career paths: Research Scientist, ML Engineer, Data Analyst, AI Product Manager, and more.

Core Pillars of Data Science

Linear algebra, calculus, probability, statistics, optimisation.	Python, SQL, Spark, cloud APIs, version control.	Business context, industry data, regulatory constraints.	Storytelling, visualisation, stakeholder presentations.

2. OVERALL LEARNING TIMELINE AT A GLANCE

Estimated time assumes 2–3 focused hours per day. Adjust based on prior experience.

Phase	Duration	Focus Area
Phase 1	6–8 Weeks	Mathematics & Statistics Foundations
Phase 2	6–8 Weeks	Python Programming & Developer Tooling
Phase 3	4–6 Weeks	Data Wrangling, Cleaning & Exploratory Analysis
Phase 4	8–10 Weeks	Core Machine Learning Algorithms
Phase 5	8–10 Weeks	Advanced ML, Ensemble Methods & Deep Learning
Phase 6	8–12 Weeks	Specialisations: NLP, Computer Vision, RL, Time Series
Phase 7	6–8 Weeks	MLOps, Model Deployment & Production Systems
Phase 8	4–6 Weeks	Business Intelligence, Reporting & Data Storytelling
Phase 9	4–6 Weeks	Big Data, Distributed Computing & Cloud Platforms
Phase 10	Ongoing	Portfolio Building, Career Strategy & Leadership

Total Estimated Duration: 12–18 months for a career-ready expert level.

■ *Prior programming/maths experience can reduce Phase 1–2 by 50%.*

3. PHASE 1 — MATHEMATICAL & STATISTICAL FOUNDATIONS

Why Maths First?

Machine learning models are mathematical constructs. Without understanding the algebra behind matrix operations, the calculus behind gradient descent, or the probability behind Bayes' theorem, you will only be a tool-user — not a problem-solver. This phase builds the intellectual foundation that separates true experts from script-followers.

1A. Linear Algebra

- Scalars, vectors, matrices and tensors — definitions and notation.
- Matrix operations: addition, multiplication, transpose, inverse.
- Dot product, cross product and geometric interpretation.
- Eigenvalues and eigenvectors — the backbone of PCA and SVD.
- Singular Value Decomposition (SVD) — image compression, recommender systems.
- Vector spaces, basis, rank, null space and linear independence.
- Norms: L1 (Manhattan), L2 (Euclidean), Frobenius — used in regularisation.

1B. Calculus & Optimisation

- Limits, derivatives and the chain rule — essential for backpropagation.
- Partial derivatives and gradients — understanding loss surface topology.
- Gradient Descent, Stochastic GD, Mini-batch GD — convergence mechanics.
- Advanced optimisers: Momentum, RMSprop, Adam — when and why to use each.
- Lagrange multipliers and constrained optimisation.
- Convexity: why convex problems are easier and how to identify them.
- Taylor series approximation — used in second-order optimisation methods.

1C. Probability & Statistics

- Sample spaces, events, axioms of probability.
- Conditional probability, Bayes' Theorem and its applications in ML.
- Probability distributions: Bernoulli, Binomial, Poisson, Normal, Exponential.
- Expectation, variance, covariance, correlation.
- Central Limit Theorem — why it enables so many statistical methods.
- Hypothesis testing: null/alternative hypotheses, p-values, Type I & II errors.
- Confidence intervals, t-tests, chi-squared tests, ANOVA.
- Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP).
- Bayesian inference — prior, likelihood, posterior, credible intervals.
- Statistical power, effect size and sample size determination.

Phase 1 Resources

Resource	Type	Best For
Mathematics for ML (Deisenroth)	Free Book	Linear algebra + probability
3Blue1Brown — Essence of LA	YouTube	Intuitive visual explanations
Khan Academy Statistics	Online Course	Statistics from scratch

StatQuest with Josh Starmer	YouTube	ML-focused stats breakdowns
Gilbert Strang — Linear Algebra	MIT OCW	Deep mathematical rigour

4. PHASE 2 — PROGRAMMING & TOOLING MASTERY

Python — The Primary Language of Data Science

Python dominates data science because of its readability, rich ecosystem, and community. You must achieve fluency — not just familiarity. Focus on writing clean, idiomatic, efficient Python before adding libraries.

2A. Core Python

Data types: int, float, str, bool, NoneType	Data structures: list, tuple, dict, set	Comprehensions: list, dict, set, generator
Functions, *args, **kwargs, closures	OOP: classes, inheritance, dunder methods	Error handling: try/except/finally
File I/O: reading/writing CSV, JSON, text	Modules and package management (pip, conda)	Decorators and context managers
Iterators, generators, lazy evaluation	Type hints and mypy basics	Virtual environments and project structure

2B. Scientific Python Stack

NumPy — Array operations, broadcasting, vectorisation — the foundation of scientific Python.

Pandas — DataFrames, Series, groupby, merge, pivot tables, time-series indexing.

Matplotlib — Low-level plotting: figures, axes, subplots, annotations, custom styling.

Seaborn — Statistical visualisation: distribution plots, heatmaps, pair plots.

SciPy — Scientific computing: statistics, signal processing, optimisation, linear algebra.

Statsmodels — Classical statistical models: OLS, GLM, time series (ARIMA), hypothesis tests.

2C. SQL & Database Fundamentals

- SELECT, WHERE, GROUP BY, ORDER BY, HAVING — query fundamentals.
- JOINs: INNER, LEFT, RIGHT, FULL OUTER, CROSS — when to use each.
- Window functions: ROW_NUMBER, RANK, LAG, LEAD, SUM OVER PARTITION.
- CTEs (Common Table Expressions) for readable complex queries.
- Subqueries, correlated subqueries, EXISTS and IN.
- Indexes, query plans and performance optimisation basics.
- Database types: PostgreSQL, MySQL, SQLite — and when to use NoSQL (MongoDB).

2D. Developer Tooling & Best Practices

- Git & GitHub: branching, pull requests, merge conflicts, semantic commits.
- Jupyter Notebooks and JupyterLab — interactive development and storytelling.
- VS Code / PyCharm — debugging, linting, formatting (Black, flake8).
- Command line / Bash scripting: navigation, pipes, grep, awk, cron jobs.
- Virtual environments: venv, conda environments, requirements.txt.
- Docker basics: containers, images, Dockerfile — for reproducible ML environments.

5. PHASE 3 — DATA WRANGLING & EXPLORATORY ANALYSIS

The 80% Rule: Why Data Engineering Matters Most

In practice, data scientists spend roughly 80% of their time acquiring, cleaning, and transforming data. Clean data is the single most important factor in model quality. No algorithm compensates for corrupted inputs.

3A. Data Acquisition

Reading CSV, Excel, JSON, Parquet, HDF5	Web scraping with BeautifulSoup & Scrapy	REST API calls with the requests library
Connecting to SQL databases with SQLAlchemy	Reading from cloud storage (S3, GCS, Azure Blob)	Streaming data with Kafka basics

3B. Data Cleaning Techniques

- Missing value detection: isnull(), info(), describe().
- Imputation strategies: mean/median/mode, forward fill, KNN imputation, MICE.
- Outlier detection: IQR method, z-scores, isolation forest, visual inspection.
- Duplicate detection and removal: drop_duplicates(), similarity matching.
- Data type coercion: strings to datetime, categorical encoding.
- String cleaning: strip, lower, regex substitution, standardisation.
- Schema validation with libraries such as Great Expectations or Pandera.

3C. Feature Engineering

- Numerical transformations: log, square root, Box-Cox, min-max, standard scaling.
- Binning/discretisation: equal-width, equal-frequency, custom bins.
- Interaction features: polynomial features, ratios, products.
- Date/time features: hour, weekday, is_weekend, time since event.
- Aggregation features: group statistics (mean, std, count) per category.
- Encoding categoricals: one-hot, ordinal, target encoding, hashing.
- Dimensionality reduction for feature creation: PCA projections.

3D. Exploratory Data Analysis (EDA)

- Univariate analysis: histograms, box plots, density curves.
- Bivariate analysis: scatter plots, correlation matrices, cross-tabs.
- Multivariate: pair plots, parallel coordinates, 3D scatter.
- Statistical summaries: central tendency, dispersion, skewness, kurtosis.
- Answering business questions with data before modelling.
- Identifying data leakage and target contamination early.

✔ Deliverable: Complete an end-to-end EDA project on a real dataset (Kaggle, UCI, or your own) and publish it as a notebook.

6. PHASE 4 — MACHINE LEARNING CORE

The Model-Building Mindset

Before writing a single line of Scikit-learn, internalise the ML workflow: problem framing → data split → baseline → iterative improvement → evaluation → interpretation. Understanding why each step exists prevents common modelling errors.

4A. The Machine Learning Workflow

- Problem framing: regression vs classification vs clustering vs ranking.
- Train/Validation/Test splits — preventing data leakage.
- Baseline models: always start simple (mean predictor, logistic regression).
- Cross-validation: k-fold, stratified k-fold, time-series split.
- Hyperparameter tuning: Grid Search, Random Search, Bayesian Optimisation.
- Model evaluation: choosing the right metric for the business problem.
- Model interpretation: SHAP, LIME, permutation importance.

4B. Supervised Learning Algorithms

Algorithm	Type	Key Hyperparameters	When to Use
Linear Regression	Regression	regularisation (α)	Continuous targets, interpretability
Ridge / Lasso / ElasticNet	Regression	alpha, l1_ratio	High-dim, correlated features
Logistic Regression	Classification	C, penalty	Binary/multi-class, interpretable
k-Nearest Neighbours	Both	k, metric	Small datasets, non-linear
Naive Bayes	Classification	var_smoothing	Text classification, fast
Support Vector Machine	Both	C, kernel, gamma	High-dim, small samples
Decision Tree	Both	max_depth, min_samples	Interpretable, rule extraction
Random Forest	Both	n_estimators, max_features	Robust general-purpose
Gradient Boosting (XGBoost/LGBM)	Both	lr, depth, subsample	Tabular data competitions

4C. Unsupervised Learning

K-Means Clustering	Hierarchical / Agglomerative	DBSCAN (density-based)
Gaussian Mixture Models	PCA — Principal Component Analysis	t-SNE — dimensionality reduction
UMAP — modern alternative to t-SNE	Autoencoders (intro)	Association Rules (Apriori)

4D. Evaluation Metrics Deep Dive

- Regression: MAE, MSE, RMSE, MAPE, R^2, Adjusted R^2.
- Classification: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, MCC.
- Ranking: NDCG, MAP, MRR.
- Calibration: reliability diagrams, Brier score, log-loss.

- Choosing metrics aligned with business cost of false positives vs negatives.

7. PHASE 5 — ADVANCED ML & DEEP LEARNING

5A. Ensemble Methods

- Bagging: variance reduction through averaging (Random Forests).
- Boosting: sequential error correction (AdaBoost, Gradient Boosting, XGBoost, LightGBM, CatBoost).
- Stacking: meta-learner trained on base-model predictions.
- Blending: holdout-set based weighted averaging.
- When to prefer each ensemble type and diversity principles.

5B. Neural Networks & Deep Learning Foundations

- Perceptron, multi-layer perceptron (MLP) and forward pass.
- Activation functions: ReLU, Sigmoid, Tanh, Leaky ReLU, GELU, Swish.
- Backpropagation step-by-step — chain rule in matrix form.
- Loss functions: MSE, Cross-entropy, Huber, Focal loss.
- Regularisation: Dropout, Batch Normalisation, Weight Decay, Label Smoothing.
- Learning rate schedules: step decay, cosine annealing, warm restarts.
- Optimisers in depth: SGD+momentum, Adam, AdamW, Lion.
- PyTorch fundamentals: tensors, autograd, custom Dataset, DataLoader, training loop.
- TensorFlow / Keras: Sequential API, Functional API, custom layers.

5C. Convolutional Neural Networks (CNN)

Convolution operation & padding	Pooling layers: max, average	LeNet, AlexNet, VGG architecture
ResNet — skip connections	Inception / GoogLeNet	EfficientNet scaling rules
Transfer Learning & fine-tuning	Data augmentation strategies	Object detection: YOLO, Faster R-CNN

5D. Recurrent Neural Networks & Sequence Models

- Vanilla RNN — vanishing gradient problem.
- LSTM: cell state, forget gate, input gate, output gate.
- GRU: simplified gating mechanism.
- Bidirectional RNNs and deep stacked RNNs.
- Sequence-to-sequence models and attention mechanism.
- Transformers: self-attention, multi-head attention, positional encoding.
- BERT, GPT architecture overview and fine-tuning approaches.
- Hugging Face ecosystem: datasets, tokenizers, model hub, Trainer API.

8. PHASE 6 — SPECIALISATIONS

Choose at least ONE deep specialisation to differentiate yourself. You can return to others after your first job.

6A. Natural Language Processing (NLP)

- Text preprocessing: tokenisation, stemming, lemmatisation, stop words.
- Bag of Words, TF-IDF, n-gram models.
- Word embeddings: Word2Vec (CBOW, Skip-gram), GloVe, FastText.
- Contextual embeddings: ELMo, BERT, RoBERTa, sentence-transformers.
- Tasks: sentiment analysis, NER, POS tagging, dependency parsing.
- Text classification, question answering, summarisation, translation.
- Large Language Models (LLMs): prompting, RAG, fine-tuning, RLHF overview.
- Vector databases: Pinecone, Weaviate, FAISS for semantic search.
- LangChain / LlamaIndex for LLM-powered applications.

6B. Computer Vision (CV)

- Image classification, object detection, instance segmentation.
- Semantic segmentation: U-Net, DeepLab.
- Generative models: GANs (DCGAN, StyleGAN), VAEs, Diffusion Models (Stable Diffusion).
- Video understanding: optical flow, action recognition.
- Medical imaging, satellite imagery, autonomous vehicles domain knowledge.
- OpenCV for classical image processing pipelines.

6C. Time Series Analysis

- Stationarity, ADF test, KPSS test, differencing.
- Classical models: AR, MA, ARIMA, SARIMA, Exponential Smoothing.
- Feature-based approaches: lag features, rolling stats with gradient boosting.
- Deep learning for time series: LSTM, TCN, N-BEATS, Temporal Fusion Transformer.
- Anomaly detection in streams: LSTM autoencoders, isolation forest, CUSUM.
- Forecasting at scale with Prophet, Darts, or NeuralForecast.

6D. Reinforcement Learning (RL)

- Markov Decision Processes: state, action, reward, policy, value function.
- Dynamic programming: policy iteration, value iteration.
- Model-free RL: Monte Carlo, TD-learning, Q-learning, SARSA.
- Deep RL: DQN, Double DQN, Dueling DQN, Policy Gradient, PPO, A3C.
- Multi-armed bandits for recommendation and A/B testing.
- OpenAI Gymnasium / Stable-Baselines3 for practical experiments.

9. PHASE 7 — MLOps, DEPLOYMENT & PRODUCTION ENGINEERING

The Production Gap: Why Models Fail After Deployment

Building a model is 20% of the work. Making it reliable, scalable, monitored, and continuously improving in production is the other 80%. MLOps bridges the gap between experimental data science and reliable software engineering.

7A. Model Deployment Approaches

REST API with FastAPI / Flask	Batch inference pipelines	Real-time streaming inference
Edge deployment (ONNX, TensorFlow Lite)	Serverless (AWS Lambda, GCF)	Gradio / Streamlit for quick demos
Docker containerisation	Kubernetes orchestration basics	

7B. ML Experiment Tracking & Versioning

- MLflow: experiments, runs, artifacts, model registry.
- Weights & Biases (W&B;): sweeps, team collaboration, dashboards.
- DVC (Data Version Control): versioning datasets and pipelines like Git.
- Feature stores: Feast, Hopsworks — sharing features across teams.

7C. ML Pipelines & Orchestration

- Scikit-learn Pipelines — chaining transformers and estimators.
- Apache Airflow — DAG-based workflow scheduling.
- Kubeflow Pipelines — ML workflows on Kubernetes.
- Prefect / ZenML / Metaflow — modern pipeline orchestration.
- CI/CD for ML: GitHub Actions, GitLab CI for automated retraining.

7D. Model Monitoring & Governance

- Data drift detection: Kolmogorov-Smirnov, Population Stability Index.
- Concept drift: monitoring prediction distributions, target drift.
- Model performance degradation alerts and automatic retraining triggers.
- Fairness and bias auditing: demographic parity, equalised odds.
- Explainability for compliance: SHAP, LIME, model cards.
- Logging, tracing and observability: Prometheus, Grafana, ELK stack.

10. PHASE 8 — BUSINESS INTELLIGENCE & DATA STORYTELLING

Technical skill without communication is a career ceiling. The best data scientists translate findings into clear decisions for non-technical stakeholders.

8A. Business Intelligence & Dashboarding

Tableau — interactive dashboards	Power BI — Microsoft BI ecosystem	Metabase / Superset (open source)
Plotly Dash — Python-based dashboards	KPI definition and tracking frameworks	A/B testing analysis and reporting

8B. Data Storytelling Principles

- The SCQA framework: Situation, Complication, Question, Answer.
- Designing for your audience — executive vs technical vs customer.
- Choosing the right chart type: comparisons, distributions, compositions, trends.
- Cognitive load reduction: Gestalt principles, pre-attentive attributes.
- Narrative structure: hook, insight, evidence, implication, recommendation.
- Presenting uncertainty: confidence intervals, error bars, scenario analysis.

8C. Analytics & Experimentation

- A/B testing design: randomisation, sample size, statistical power.
- Multi-variate testing (MVT) and factorial designs.
- Causal inference: difference-in-differences, propensity score matching.
- Instrumental variables and regression discontinuity designs.
- North Star metrics, OKRs and building a data-driven culture.

11. PHASE 9 — BIG DATA & CLOUD PLATFORMS

When your data exceeds RAM, you need distributed computing. Cloud fluency is now expected at senior levels.

9A. Big Data Fundamentals

- Hadoop ecosystem: HDFS, MapReduce, YARN — historical context.
- Apache Spark: RDDs, DataFrames, Spark SQL, PySpark, MLlib.
- Spark Streaming and Structured Streaming for real-time pipelines.
- Apache Kafka: producers, consumers, topics, partitions, offsets.
- Apache Flink for stateful stream processing.
- Data lakehouse architecture: Delta Lake, Apache Iceberg, Apache Hudi.

9B. Cloud Platforms for Data Science

AWS Services	GCP Services	Azure Services
SageMaker (end-to-end ML)	Vertex AI (end-to-end ML)	Azure ML Studio
S3 (object storage)	Cloud Storage (GCS)	Azure Blob Storage
Redshift (data warehouse)	BigQuery (serverless DW)	Azure Synapse Analytics
EC2, Lambda, ECS, EKS	Compute Engine, Cloud Run	Azure VMs, AKS
Glue (ETL), Athena (SQL)	Dataflow (Apache Beam)	Azure Data Factory

9C. Data Engineering Fundamentals

- ETL vs ELT pipeline design and when to use each.
- dbt (data build tool) for SQL-based transformations.
- Data quality frameworks: Great Expectations, Soda Core.
- Data catalogue and metadata management: Apache Atlas, DataHub.
- Data modelling: star schema, snowflake schema, OBT (One Big Table).

12. PHASE 10 — PORTFOLIO, CAREER & LEADERSHIP

Your GitHub is your resume. Your projects speak louder than certifications.

10A. Portfolio Project Ideas by Level

Level	Project Idea	Skills Demonstrated
Beginner	Titanic / House Price Prediction	EDA, cleaning, baseline ML
Beginner	Customer Churn Analysis Dashboard	SQL, Pandas, Tableau
Intermediate	Recommender System (MovieLens)	Collaborative filtering, matrix factorisation
Intermediate	NLP Sentiment & Topic Modelling	BERT fine-tuning, LDA, EDA
Intermediate	Time Series Demand Forecasting	ARIMA, LightGBM, evaluation
Advanced	End-to-End MLOps Pipeline	Docker, MLflow, FastAPI, Airflow
Advanced	Computer Vision Detection System	YOLOv8, custom dataset, deployment
Advanced	LLM-Powered RAG Application	LangChain, vector DB, evaluation
Advanced	Causal Impact Business Study	DiD, PSM, A/B experimentation

10B. Interview Preparation Strategy

- SQL interviews: practice 50+ LeetCode / StrataScratch medium-hard problems.
- ML theory: explain bias-variance tradeoff, regularisation, ensemble methods clearly.
- Statistics: hypothesis tests, A/B testing design, Bayesian vs frequentist debate.
- Coding: implement algorithms from scratch (k-means, linear regression, backprop).
- Case studies: practice the full ML pipeline for a business problem in 45 minutes.
- Behavioural: STAR method stories about data-driven decisions and impact.

10C. Career Tracks & Progression

Role	Key Focus	Progression Path
Data Analyst	SQL, BI, reporting	→ Senior Analyst → Data Scientist
Data Scientist	ML modelling, statistics	→ Senior DS → Staff DS → Principal
ML Engineer	Deployment, MLOps, scale	→ Senior MLE → ML Platform Engineer
Research Scientist	Novel algorithms, papers	→ Senior Researcher → Research Lead
Data Engineer	Pipelines, data platform	→ Senior DE → Data Architect
AI Product Manager	Strategy, roadmap	→ Director of AI Product

10D. Senior-Level Skills & Leadership

- Mentoring junior data scientists and conducting code reviews.
- Defining team-wide standards for experimentation, modelling and deployment.
- Communicating with C-suite on ROI, risk and strategic AI direction.
- Building and owning ML platforms that scale across the organisation.

- Contributing to open source and publishing research or technical blog posts.
- Staying current: reading arXiv papers, attending NeurIPS / ICML / KDD.

13. CURATED LEARNING RESOURCES

Books

Resource	Type	Best For
Hands-On ML (Géron)	Book	Scikit-learn, Keras, TensorFlow — practical
Deep Learning (Goodfellow et al.)	Book	Mathematical foundations of DL
The Elements of Statistical Learning	Free PDF	Statistical ML theory
Designing ML Systems (Huyen)	Book	Production-ready ML thinking
Python for Data Analysis (McKinney)	Book	Pandas from the creator
Feature Engineering Bookcamp	Book	Practical feature techniques

Online Courses & Platforms

Resource	Type	Best For
fast.ai (Practical DL)	Free MOOC	Top-down DL with PyTorch
Coursera ML Specialisation (Ng)	MOOC	Foundational ML with Python
DeepLearning.AI courses	MOOC	NLP, MLOps, CV specialisations
Kaggle Learn	Free Micro	Hands-on short courses
full stack deep learning	Free MOOC	Production ML engineering
CS229 / CS231n / CS224n (Stanford)	Free OCW	Research-level courses

14. COMMON MISTAKES & HOW TO AVOID THEM

✕ Skipping maths
Rushing into Scikit-learn without understanding the maths leads to black-box thinking and inability to debug or innovate.

✕ Tutorial purgatory
Watching courses endlessly without building original projects. After each section, build something real.

✕ Ignoring data quality
Spending 90% of time on modelling and 10% on data. Reverse this ratio.

✕ Overfitting your resume
Listing every library rather than demonstrating project impact and measurable results.

✕ Choosing wrong metrics
Optimising accuracy on imbalanced data leads to deceptive results. Always question metric choice.

✕ Neglecting communication
Being unable to explain a model to a non-technical manager is a career barrier at senior levels.

✕ Solo learning only
Not joining communities (Kaggle, local meetups, open-source contributions) slows your growth.

✕ Specialising too early
Going deep into deep learning before mastering classical ML means weak foundations.

✕ Ignoring deployment
Building models that live only in notebooks is not production data science.

✕ Not reading papers
At the expert level, breakthroughs come from research. Develop the habit of reading arXiv papers weekly.

15. FINAL EXPERT CHECKLIST & CLOSING THOUGHTS

Use this checklist to assess your progression to expert level:

■ Mathematics

- ■ Can derive gradient descent and explain its convergence conditions
- ■ Understand eigendecomposition and its role in PCA
- ■ Can design and interpret a hypothesis test end-to-end

■ Programming

- ■ Write clean, tested, documented Python code
- ■ Comfortable with SQL window functions and performance tuning
- ■ Use Git proficiently in a team environment

■ Machine Learning

- ■ Implement linear regression and logistic regression from scratch
- ■ Explain bias-variance tradeoff with a concrete example
- ■ Tune XGBoost/LightGBM for a production dataset

■ Deep Learning

- ■ Build and train a CNN or Transformer architecture from scratch
- ■ Fine-tune a pre-trained LLM for a downstream task
- ■ Debug training instability (exploding/vanishing gradients)

■ MLOps

- ■ Deploy a REST API serving a trained model in Docker
- ■ Set up experiment tracking with MLflow or W&B;
- ■ Implement a basic data drift monitoring system

■ Communication

- ■ Present a data science project to non-technical stakeholders
- ■ Write a clear technical blog post or case study
- ■ Design an A/B test and explain the results to a product team

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Closing Thoughts

Becoming a Data Science expert is not a destination — it is a continuous journey. The field evolves rapidly, and the greatest practitioners combine deep technical knowledge with intellectual curiosity, rigorous scepticism, and genuine empathy for the people their models affect. Build with purpose, measure with honesty, and communicate with clarity.

The roadmap above is your compass, not your cage. Adapt it to your context, your industry, and your ambitions. The most important step is always the next one.

— End of Data Science Expert Roadmap —